



NILUS LAB

Reversing Cardiac Ageing Without Oncogene Risk

Generative cellular reprogramming therapies targeting cardiomyopathy and heart failure.
Powered by the Zenith v28.0 GOLD single-cell foundation model.

DEEP BIO ACCELERATOR - LAUNCH TRACK INTERVIEW PRESENTATION

CLINICAL RISK & LIMITS

Cardiac Senescence & Reprogramming Limits

- **64M+ Patients Globally:**
Heart failure is driven by cellular senescence, loss of adult sarcomere structures, and chromatin degradation.
- **Oncogenic Risk (OSKM):**
Classical Yamanaka factors trigger dedifferentiation and cell drift, creating high risks of teratoma and tumor formation.
- **Delivery & Specificity Limits:**
Existing therapies cannot precisely target therapeutic chromatin states, leading to unsafe, non-specific changes.

ZENITH PLATFORM SOLUTION

Safe, Target-Specific Rejuvenation

- **In-Silico Safety Gating:**
Dual-threshold gates screen out oncogenes and restrict cellular dedifferentiation past functional limits.
- **-11.9y Epigenetic Reset:**
Validated age clock trained on real HCA data achieves 11.9 years of age reversal in mature cells.
- **Transient Cooperativity:**
Optimized expression windows (e.g., 2h ON) regulate chromatin accessibility without altering cell identity.

1 scVI Generative Core

- Trained on 500,000 human cardiac cells from HCA + PERIHEART.
- Maps cellular reprogramming dynamics across 4,908 highly variable gene dimensions.
- Generates in-silico cell state representations to predict the effects of transcription factor interventions.

2 Safety Auditing

- Oncogene Blacklist: Screens candidate genes to eliminate tumor-forming risks.
- Dedifferentiation Ceiling: Restricts cellular state changes past strict safety boundaries.
- Prevents tissue identity loss during the rejuvenation process.

3 AF3 & Dosage Audit

- AlphaFold 3 Valency: Models transcription factor binding affinities (e.g. POU5F1-SOX2 heterodimer).
- Bayesian Dosage Networks: Predicts expression patterns to find optimal dose-response curves.
- Design: Prescribes transient expression windows for in-vitro translation.

MARKET POSITIONING

Target Market & Indication Strategy

- TAM: \$50B longevity and age-induced cardiovascular indications.
- Indications: Age-induced cardiomyopathy, vascular wall stiffness, and ischemic heart failure.
- Business Model: Out-license verified in-silico gene cocktails to pharma partners, accelerating discovery times from 5 years to 3 weeks.

ACCELERATOR SELECTION & INTERVIEW STATUS

Selected for founder video interview with the Deep Bio Accelerator (Launch Track).

Preparing video call with the accelerator selection committee to review clinical strategy, milestones, and co-founder dynamics.

\$0M - \$2M

LAUNCH TRACK FUNDING BAND

-11.9 Years

EPIGENETIC AGE RESET VALIDATED

THE TEAM

Team Chemistry & Dynamics

- Interdisciplinary Founders: Unifies expertise across software engineering, single-cell variational autoencoders, and structural molecular biology.
- Long-standing Research Partnership: The co-founders have a history of collaborative research in single-cell transcriptomics.
- Interview Preparedness: Scheduled video call to align on Zenith's technical architecture, safety thresholds, and program goals.

DEVELOPMENT ROADMAP

Key Execution Milestones



Phase 1: Program & Interview Prep

Complete the Deep Bio founder interview, finalize research targets, and configure entities.



Phase 2: Preclinical In-Vitro Validation

Synthesize predicted cocktails and run safety assays on primary patient cells.



Phase 3: Seed Round & IND Enabler

Raise a \$1.5M seed round to compile preclinical safety data and initiate IND filing.